



Cloud-Native SIG present

Kubernetes for Research Infrastructure: Core Concepts and Practical Use Cases

20 August 2026, ACIT Hub Webinar

Agenda

1. Introductions
2. Introduction to Kubernetes
3. Use case 1: Lab in the Cloud
4. Use case 2: Jupyter Notebooks
5. Use case 3: Argo CD and Workflows
6. Summary
7. Q&A

Introductions



Who We Are

This webinar is run by the **Cloud-Native Special Interest Group**, a new community of RSEs/dRTPs for cloud-native technologies in research software and digital infrastructure.

Get involved at <https://cloudnative-sig.ac.uk/>

Current committee includes; Laura Shemilt¹, Alex Lubbock², Piper Fowler-Wright², Lewis Sampson³, and Tibor Auer⁴.

We are joined today by Dr Christopher Green³, Sys Admin in Scientific Computing Department

¹BioFAIR, ²Rosalind Franklin Institute, ³UKRI STFC DAFNI, ⁴University College London



The SIG so far

We have been;

- Building networks through conference attendance
- Running Kubernetes introduction workshops for RSE/dRTPs
- Sharing knowledge of Kubernetes in the DRI community through CAKE fellowship
- Progressing towards being formalised as a RSE Society.

If you are interested

Our workshops are available via our GitHub pages

- [Deploying a Webservice with Kubernetes](#)
- [Pocket sized Kubernetes](#): For this one, to recreate with your own hardware, see the [extra reading section](#).

Introduction to Kubernetes

What is Kubernetes (k8s)

Kubernetes is an open-source container orchestration platform to automate the deployment, scaling & management of containerised applications across groups of machines.

Kubernetes vs Docker Compose

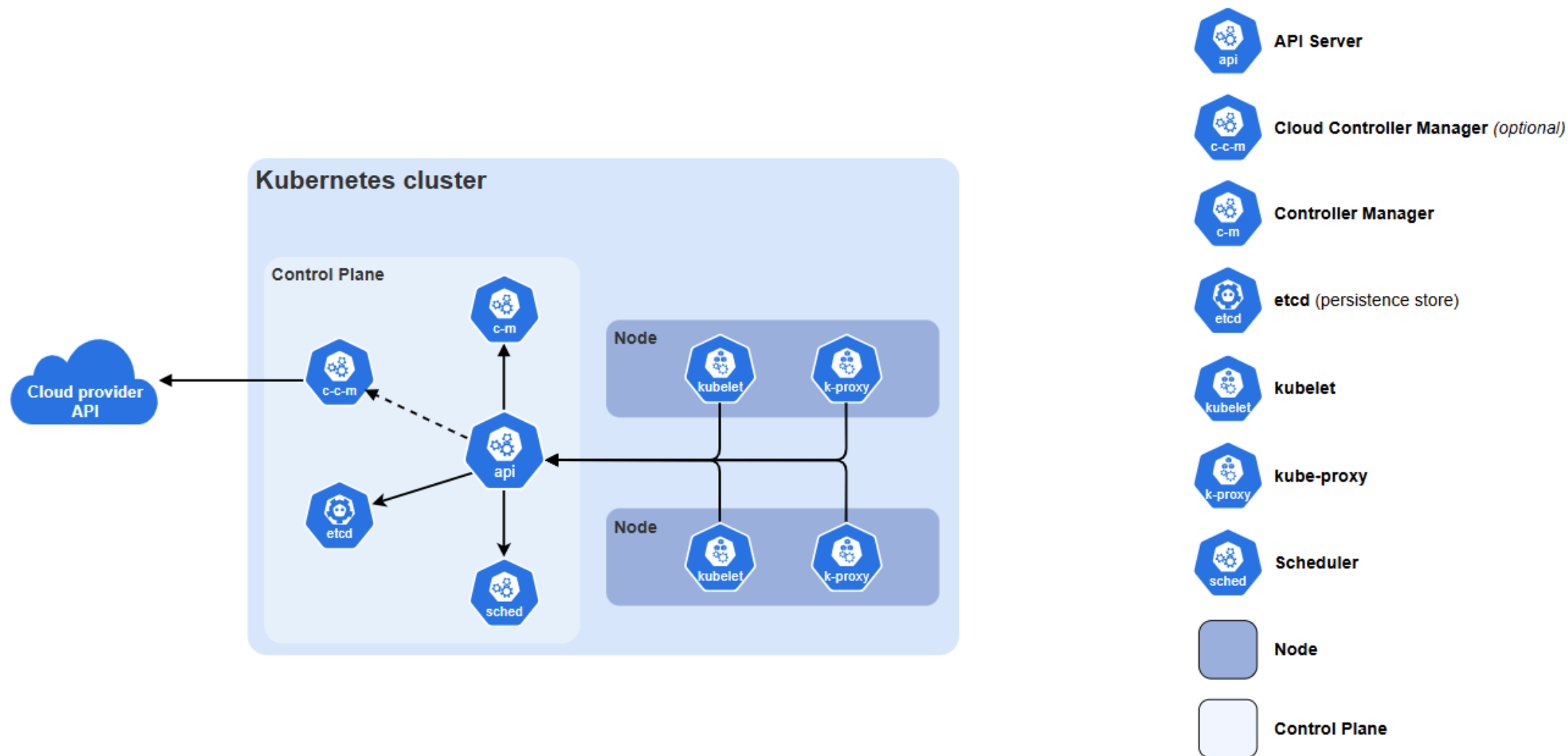
- **Docker:** run apps from built images in containers (sandboxed environments)
- **Docker Compose:** groups of containers with networking and storage on a single host
- **Kubernetes:** Orchestration pipeline for multi-host, multi-container applications.

K8s allows complex containerised applications to run across multiple hosts in a cluster with powerful automation and management features.

Key Components

- **Node:** Physical or virtual machine
- **Control Plane:** Brains of cluster, acts based on current cluster state vs target state
- **Worker:** Run containerised applications using a container runtime.
- **Pod:** Smallest deployable unit—one or more containers sharing storage/network
- **Deployment:** Target state of identical pods serving an app
- **Service:** Network endpoint for pods

Architecture Overview



How Kubernetes Works

Benefits

- **Declarative** Configuration. **GitOps**: Infrastructure as code, workflow managers, e.g., [Flux](#), [ArgoCD](#)
- General **resource management** including networking and security
- Ability to **scale clusters** to include additional cloud resources.
- Offer High Availability (HA) with **Self-Healing**.

Example: On *node failure* Kubernetes distributes workload to other nodes

How Kubernetes Works

Difficulties

- Investing in an entirely new infrastructure
 - Research and Investigation
 - Migration
 - Upskilling
- Fast paced update cycle and dependency chains

Kubernetes uses in Research Computing

Use if you need to manage multiple containerised services across different machines, or for:

- Automatic workload scaling based on demand
- Rolling updates with minimal downtime
- Self-healing resilient systems with high-availability
- Portability between on-prem and cloud infrastructure

Kubernetes not used in Research Computing

Adds complexity and not typically suitable for:

- Simple container applications on a single host (use Docker Compose)
- HPC job management (use SLURM)
- Large-scale parallel filesystems (e.g., Lustre)
- Services offered by a cloud provider or technology you are already invested in

Use cases

Wrap-up

Cloud Native SIG

This webinar was brought to you by the Cloud-Native SIG and CAKE, with support from the Software Sustainability Institute

You can join the CNSIG

- We are looking to grow the SIG so please join our Jisc mailing list for future updates.
 -  cloudnative-sig@jiscmail.ac.uk
 -  cloudnative-sig.ac.uk
- Or if you are deeply interested Kubernetes and Cloud-Native technologies join our committee. Contact any of us on email or LinkedIn to show interest



Next up

Join the CNSIG at RSECon 2026!

We will be running the *Deploying a Webservice with Kubernetes* workshop on Thursday the 10th September. If you're going to be at the conference feel free to come along, and catch the CNSIG throughout the event.

Stay in touch

Feel free to contact the SIG after the webinar using the websites "Contact Us" page.

<https://cloudnative-sig.ac.uk/pages/contact-us.html>

**Thank you for listening, any
questions?**